

Adithya Hegde

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github.com/TheMonocledHamster

Education

Georgia Institute of Technology, MS in Computer Science - Computing Systems Expected May 2028

The National Institute of Engineering, BE in Computer Science and Engineering Jun 2023

Experience

Software Engineer II, JPMorgan Chase & Co. Jan 2025 – Jul 2026

- Owned the technical design and implementation of a new Spark-based data platform replacing legacy processing systems, from architecture through production rollout.
- Redesigning platform execution to eliminate Databricks-DynamoDB state synchronization, removing a fragile distributed dependency and improving throughput by up to 80%.
- Reduced runtime of a complex ~60-operation Spark DAG processing 1-10M records from ~10 minutes to ~1.5 minutes through logical-plan optimization and data-dependent DAG construction that eliminated redundant execution stages.
- Redesigning partitioning and parallelism to mitigate shuffle bottlenecks, reducing cluster compute costs by ~40% without performance degradation.
- Optimized table partitioning using Spark execution plans and Databricks query profiles, increasing data skipping from 65% to 98% and reducing dashboard refresh latency from ~5 minutes to ~10 seconds.

Software Engineer I, JPMorgan Chase & Co. Jul 2023 – Dec 2024

- Profiled memory growth of a Polars-based transformation running in AWS Lambda's constrained runtime; introduced lazy evaluation and explicit object lifecycle control, reducing peak memory allocation by 60% and stabilizing execution latency.
- Brought up a multi-node ingestion system on Kubernetes (EKS + NiFi), implemented Airflow orchestration for distributed workflows, and operated cross-service data movement pipelines in production.
- Earned accelerated promotion to Software Engineer II for ownership of distributed data infrastructure and production performance engineering.

Software Engineer Intern, JPMorgan Chase & Co. Jan 2023 – Jun 2023

- Implemented a schema-driven validation tool generating record-level checks from external specifications, eliminating iterative reprocessing and reducing failure diagnosis time by 95%.

Publications

COUNSEL - Cloud Resource Configuration Management using Deep Reinforcement Learning May 2023

Proposed a DRL-based framework for cloud resource configuration using queueing-based service-chain modeling, and evaluated system performance under simulated workloads. Artifacts available on GitHub ([Counsel](#)).

Adithya Hegde, Sameer G. Kulkarni, Abhinandan S. Prasad
[10.1109/ccgrid57682.2023.00035](https://doi.org/10.1109/ccgrid57682.2023.00035) (IEEE/ACM CCGRID 2023)

Skills

Languages: Python, Java, SQL; Familiarity with Rust, C++

Systems & Data: Spark, Databricks, Airflow, Kubernetes, AWS, Linux